



Travel AI



Le Vietnam



Le Vietnam - les paysages



Le Vietnam - la bouffe



Le Vietnam - la culture



Le Vietnam - ...la bouffe



Le Vietnam - ...???



Preparer ses vacances

Mes options :

- Passer des heures sur internet à trouver les meilleurs plans
- Demander à un ami qui a déjà été sur place
- Demander à un agent de voyage
- Déléguer à une ou plusieurs IA

Mes premiers tests

...Ce nouveau reflexe



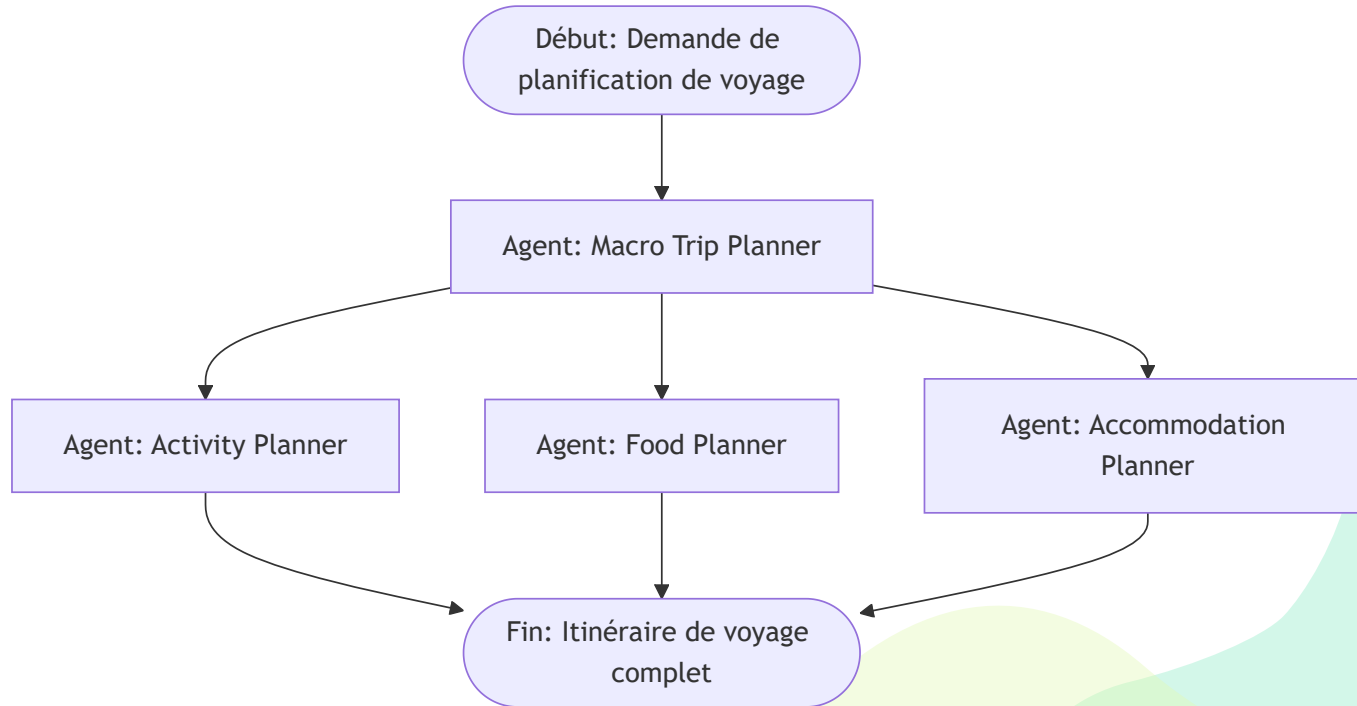
Le poto

1/58

Les problèmes de cette solution

- 🤖 La réponse manque de détails
- 🎨 Peu de personnalisation à moins de faire un grand nombre d'aller retour
- 🍷 Pas de proposition de lieux d'hébergement, de restauration, de prix...
- 🌐 Des données plus ou moins récente

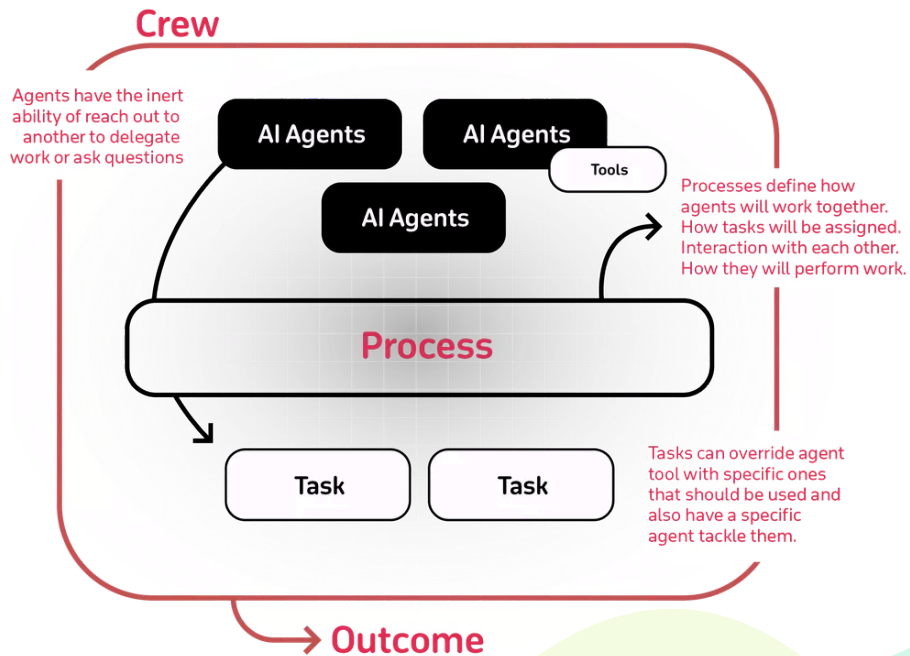
Le début du travail forcé



L'outillage



L'outillage



Les agents

```
macro_trip_planner:  
  role: Macro trip planner  
  goal: >  
    Create a comprehensive travel plan that includes all aspects of the trip, from departure to return  
  backstory: >  
    You are an expert in travel planning, with a deep understanding of logistics, budgeting, and travel  
    You can create macroscopic itineraries with 2 or 3 main cities which are central points of tourism
```

Les tasks

macro_planning_task:

description: >

Create a comprehensive travel plan that includes all aspects of the trip, from departure to return. The plan should include details about the itinerary and activities. The plan should NOT include details about the accommodations and activities. Focus on creating a macroscopic itinerary with 2 or 3 main cities that are central points of tourism, allowing access to the most interesting tourist elements.

Traveler's information:

- origin: {original_country}
- destination: {destination_country}

expected_output: >

A detailed travel plan that includes the following information:

- Departure and return dates
- Itinerary with 2 or 3 main cities
- Transportation details (flights, trains, etc.)
- Estimated travel times between cities
- Budget considerations (if applicable)
- Any special notes or considerations for the traveler

agent: macro_trip_planner

L'équipe

```
@CrewBase
class TravelCrew():

    @before_kickoff
    def prepare_inputs(self, inputs): ...

    @agent
    def macro_trip_planner(self) → Agent: ...

    @task
    def macro_planning_task(self) → Task: ...

    @crew
    def crew(self) → Crew:
        return Crew(
            agents=self.agents,
            tasks=self.tasks
        )
```

Les agents

```
macro_trip_planner:
  role: Macro trip planner
  goal: >
    Create a comprehensive travel plan that includes all aspects of the trip, from departure to return
  backstory: >
    You are an expert in travel planning, with a deep understanding of logistics, budgeting, and travel
    You can create macroscopic itineraries with 2 or 3 main cities which are central points of tourism

activity_planner:
  role: Activity planner
  goal: >
    Research and find cool things to do at the destination, including activities and events that match
  backstory: >
    You are skilled at creating personalized itineraries that cater to the specific preferences and de

food_planner:
  role: Food planner
  goal: >
    Research and find places to eat at the destination, including restaurants, cafes, and local food ex
```

Les tasks

macro_planning_task:

description: >

Create a comprehensive travel plan that includes all aspects of the trip, from departure to return.
The plan should include details about the itinerary and activities.
The plan should NOT include details about the accommodations and activities.
Focus on creating a macroscopic itinerary with 2 or 3 main cities that are central points of tourism, allowing access to the most interesting tourist elements.

Traveler's information:

- origin: {original_country}
- destination: {destination_country}

expected_output: >

- A detailed travel plan that includes the following information:
- Departure and return dates
 - Itinerary with 2 or 3 main cities
 - Transportation details (flights, trains, etc.)

Démo time



Le résultat – Macro Planning

****Comprehensive Travel Plan from France to Vietnam****

****Departure and Return Dates:****

- ****Departure from France:**** March 15, 2024
- ****Return to France:**** March 30, 2024

****Itinerary:****

- ****City 1: Hanoi****
 - **Duration:** March 16 - March 20
 - **Highlights:** Explore the Old Quarter, Hoan Kiem Lake, Ho Chi Minh Mausoleum, and take a day trip to Ha Long Bay.
- ****City 2: Da Nang/Hoi An****
 - **Duration:** March 21 - March 24
 - **Highlights:** Visit the Marble Mountains, relax on My Khe Beach, enjoy the charm of Hoi An's Ancient Town.

Le résultat – Activity Planning

****Comprehensive Itinerary for Vietnam Trip****

****City 1: Hanoi (March 16 – March 20)****

1. **March 16: Explore Hanoi's Old Quarter**

- ****Activity:**** Walking tour through Hanoi Old Quarter
- ****Location:**** Old Quarter, Hanoi
- ****Description:**** Discover the bustling streets filled with French colonial architecture, bustling
- ****Why Suitable:**** Perfect for immersing in local culture and trying authentic Vietnamese street fo
- ****Potential Reviews:**** Rated 4.5/5 on TripAdvisor for its vibrant ambience and historical signific

2. **March 17: Ho Chi Minh Mausoleum & Hoa Lo Prison**

- ****Activity:**** Visit Ho Chi Minh Mausoleum and Hoa Lo Prison Museum
- ****Location:**** Ba Dinh, Hanoi
- ****Description:**** Learn about Vietnam's history with a visit to the revered Mausoleum of Ho Chi Min
- ****Why Suitable:**** Offers a poignant view of Vietnam's modern history and significant figures.
- ****Potential Reviews:**** Highly rated on Google (4.6/5) for insightful historical context.

Le résultat – Food Planning

Here is a list of recommended places to eat each day of the trip in Vietnam, tailored to the traveler's

****Hanoi (March 16 - March 20)****

1. ****March 16: Hanoi Old Quarter****

- ****Place:**** Bun Cha Ta
- ****Location:**** 21 Nguyen Huu Huan, Hoan Kiem, Hanoi
- ****Description:**** A favorite spot for Bun Cha, a Vietnamese dish of grilled pork and noodles.
- ****Why Suitable:**** Offers an authentic taste of Hanoi's street food culture.
- ****Potential Reviews:**** Rated 4.5/5 on TripAdvisor for its flavorful dishes and wonderful atmosphere.

2. ****March 17: Ba Dinh, Hanoi****

- ****Place:**** KOTO Restaurant
- ****Location:**** 59 Van Mieu, Dong Da, Hanoi
- ****Description:**** A social enterprise that provides hospitality training to underprivileged youth.
- ****Why Suitable:**** Combines delicious local cuisine with a meaningful cause.
- ****Potential Reviews:**** Highly rated for both purpose and food quality (4.6/5 on Google).

3. ****March 18: Halong Bay Day Trip****

Le résultat – Accommodation Planning

****Recommended Accommodations for Vietnam Trip****

****Hanoi (March 16 – March 20)****

1. **March 16 – 20: La Siesta Hotel & Spa**

- ****Location:**** 94 Ma May, Hoan Kiem district, Hanoi
- ****Description:**** Situated in the heart of Hanoi's Old Quarter, this hotel offers a charming blend of traditional Vietnamese architecture and modern amenities.
- ****Why Suitable:**** Perfectly located for exploring Hanoi's cultural sites, offering rejuvenating spa treatments.
- ****Potential Reviews:**** Highly praised for exceptional service and ambiance, rated 4.7/5 on Booking.com.

****Da Nang/Hoi An (March 21 – March 24)****

1. **March 21 – 24: Anantara Hoi An Resort**

- ****Location:**** 1 Pham Hong Thai Street, Hoi An
- ****Description:**** Nestled along the Thu Bon River, this resort offers riverside villas surrounded by lush gardens.
- ****Why Suitable:**** Provides a tranquil escape with easy access to cultural hotspots, perfect for relaxation.
- ****Potential Reviews:**** Renowned for its location and hospitality, rated 4.8/5 on TripAdvisor.

****Ho Chi Minh City (March 25 – March 29)****

Bilan intermediaire

- On obtiens des **informations plus précise**
- On a des données specifique pour chaque poste :
 - Activité
 - Nourriture
 - Hébergement
 - ...
- Les agents on profité les uns les autres de leurs résultats
- !? Aucune recherche sur internet n'a été faite 🤔
- Résultat fournis par défaut en markdown

Formatter les outputs

Avant d'améliorer les résultats, formatons les.

Utilisation de la librairie **Pydantic** qui permet de créer et valider des modèle de données en python.

```
from pydantic import BaseModel

class City(BaseModel):
    name: str
    transport_to_come: str
    transport_price_min: float
    transport_price_max: float

class MacroPlanning(BaseModel):
    ordered_cities: list[City] = []
    travel_considerations: list[str] = []
    local_currency: str = ""
    current_exchange_rate: float = 0.0
```

Formatter les outputs

Avant d'améliorer les résultats, formatons les.

Utilisation de la librairie **Pydantic** qui permet de créer et valider des modèle de données en python.

```
from pydantic import BaseModel

class Food(BaseModel):
    name: str
    description: str
    address: str
    location: str
    price_min: float
    price_max: float
    review_score_on_5: float

class FoodList(BaseModel):
    foods: list[Food]
```

Formatter les outputs

Avant d'améliorer les résultats, formatons les.

Utilisation de la librairie **Pydantic** qui permet de créer et valider des modèle de données en python.

```
from pydantic import BaseModel

class Activity(BaseModel):
    name: str
    description: str
    duration: int
    location: str
    price_min: float
    price_max: float
    review_score_on_5: float

class ActivityList(BaseModel):
    activities: list[Activity]
```

Formatter les outputs

Avant d'améliorer les résultats, formatons les.

Utilisation de la librairie **Pydantic** qui permet de créer et valider des modèle de données en python.

```
from pydantic import BaseModel

class Accommodation(BaseModel):
    name: str
    description: str
    location: str
    price_min: float
    price_max: float
    review_score_on_5: float

class AccommodationList(BaseModel):
    accommodations: list[Accommodation]
```

Démo time



Le résultat – Macro Planning

```
{
  "ordered_cities": [
    {
      "name": "Hanoi",
      "transport_to_come": "Flight from France to Noi Bai International Airport, Hanoi",
      "transport_price_min": 600.0,
      "transport_price_max": 1000.0
    },
    {
      "name": "Da Nang",
      "transport_to_come": "Domestic flight from Hanoi to Da Nang International Airport",
      "transport_price_min": 30.0,
      "transport_price_max": 80.0
    },
    {
      "name": "Ho Chi Minh City",
      "transport_to_come": "Domestic flight from Da Nang to Tan Son Nhat International Airport, Ho Chi",
      "transport_price_min": 30.0,
      "transport_price_max": 80.0
    }
  ]
}
```

Le résultat – Activity Planning

```
{
  "activities": [
    {
      "name": "Hanoi Old Quarter Walking Tour",
      "description": "Explore the bustling streets of Hanoi's Old Quarter, known for its unique architecture and street food.",
      "duration": 4,
      "location": "Hanoi",
      "price_min": 15.0,
      "price_max": 30.0,
      "review_score_on_5": 4.8
    },
    {
      "name": "Halong Bay Cruise",
      "description": "Embark on a scenic cruise through Halong Bay, a UNESCO World Heritage site, famous for its limestone karsts and emerald waters.",
      "duration": 8,
      "location": "Halong Bay, near Hanoi",
      "price_min": 80.0,
      "price_max": 150.0,
      "review_score_on_5": 4.7
    }
  ]
}
```

Le résultat – Food Planning

```
{
  "foods": [
    {
      "name": "Bún Chả Đặc Kim",
      "description": "A popular spot for the traditional Hanoi dish, Bún Chả, which includes grilled pork",
      "address": "1 Hàng Mành, Hoàn Kiếm, Hà Nội",
      "location": "Hanoi",
      "price_min": 2.0,
      "price_max": 5.0,
      "review_score_on_5": 4.5
    },
    {
      "name": "The Note Coffee",
      "description": "A charming café known for its walls decorated with thousands of colorful sticky notes",
      "address": "64 Lương Văn Can, Hoàn Kiếm, Hà Nội",
      "location": "Hanoi",
      "price_min": 2.0,
      "price_max": 4.0,
      "review_score_on_5": 4.6
    }
  ]
}
```

Le résultat – Accommodation Planning

```
{
  "accommodations": [
    {
      "name": "Sofitel Legend Metropole Hanoi",
      "description": "A historic luxury hotel in the heart of Hanoi, known for its French colonial grandeur",
      "location": "15 Ngô Quyền, Hoàn Kiếm, Hà Nội",
      "price_min": 250.0,
      "price_max": 450.0,
      "review_score_on_5": 4.8
    },
    {
      "name": "Hanoi La Siesta Hotel & Spa",
      "description": "A charming boutique hotel situated in the Old Quarter, offering a blend of modern comfort and traditional Hanoi architecture",
      "location": "94 Mã Mây, Hoàn Kiếm, Hà Nội",
      "price_min": 80.0,
      "price_max": 140.0,
      "review_score_on_5": 4.7
    },
    {

```

Bilan intermediaire

- On a des données specifique pour chaque poste
- Résultat fournis en data mapping exploitable
- Les agents on malgré tout profité les uns les autres de leurs résultats
- **!?** Aucune recherche sur internet n'a été faite 🤔

Bon ça vient cette recherche ?

Il est temps d'ajouter un peu de dynamisme dans tout ça.

Pour ça **crewAI** integre la notion de tools.

Un tool crewAI c'est quoi ?

- Un outil qui va permettre d'effectuer une tâche spécifique
- Développé en python
- Il prend un objet en input et renvoi un output à destination du LLM
- Les tools sont exécuté en local, le LLM n'a pas un accès direct à la donnée

Exemple de tools existant

- **Web Search:** Effectue une recherche sur different moteur de recherche et renvoie les X premiers résultats
- **Web Scraping:** Permet de scraper une page web et d'en extraire les données
- **API Call:** Permet d'effectuer un appel API et de renvoyer les données
- **File Search:** Permet de rechercher un fichier sur le systeme de fichier et de renvoyer son contenu
- **Code interpreter:** Permet d'exécuter du code python et de renvoyer le résultat
- **Image generation:** Permet de générer une image à partir d'un prompt
- **Data RAG:** Permet d'adosser une source de données à un LLM et de lui permettre d'y accéder
- **SQL RAG:** Permet au LLM de faire des requête SQL en langage naturel
- ...

Mais c'est méga dur à implementé non ?

```
@CrewBase
class TravelCrew():

    @agent
    def macro_trip_planner(self) → Agent:
        return Agent(
            config=self.agents_config['macro_trip_planner'],
            verbose=True,
            tools=[
                SerperDevTool(),
                ScrapeWebsiteTool()
            ]
        )
```



Démo time



Le résultat – Macro Planning

```
{
  "ordered_cities": [
    {
      "name": "Ho Chi Minh City",
      "transport_to_come": "Flight from Paris",
      "euro_transport_price_min": 500.0,
      "euro_transport_price_max": 900.0
    },
    {
      "name": "Hanoi",
      "transport_to_come": "Domestic Flight from Ho Chi Minh City",
      "euro_transport_price_min": 30.0,
      "euro_transport_price_max": 70.0
    }
  ],
  "travel_considerations": [
    "Plan for potential layovers when flying from France to Vietnam.",
    "Check the local COVID-19 travel restrictions and requirements for 2025.",
    "Keep in mind that the Vietnamese Dong is the local currency, so exchange money beforehand."
  ]
}
```

Le résultat – Activity Planning

```
{
  "activities": [
    {
      "name": "Tet Nguyen Dan (Lunar New Year)",
      "description": "Vietnam's most significant holiday that celebrates the lunar new year through fan",
      "duration": 3,
      "location": "Nationwide, particularly in Ho Chi Minh City and Hanoi",
      "local_currency_price_min": 0.0,
      "local_currency_price_max": 0.0,
      "euro_currency_price_min": 0.0,
      "euro_currency_price_max": 0.0,
      "review_score_on_5": 5.0
    },
    {
      "name": "Tet Nguyen Tieu (Lantern Festival)",
      "description": "Marks the end of Tet celebrations with vibrant lantern displays and community ac",
      "duration": 1,
      "location": "Ho Chi Minh City and Hanoi",
      "local_currency_price_min": 0.0,
```

Le résultat – Food Planning

```
{
  "foods": [
    {
      "name": "Pho 2000",
      "description": "Known for serving arguably the best Pho in Ho Chi Minh City, this restaurant has",
      "address": "1-3 Phan Chu Trinh, Ben Thanh, District 1, Ho Chi Minh City",
      "location": "Ho Chi Minh City",
      "local_currency_price_min": 60000.0,
      "local_currency_price_max": 100000.0,
      "euro_currency_price_min": 2.08,
      "euro_currency_price_max": 3.47,
      "review_score_on_5": 4.7
    },
    {
      "name": "Banh Mi Huynh Hoa",
      "description": "Famous for its delicious Banh Mi, a Vietnamese sandwich filled with savory ingred",
      "address": "26 Le Thi Rieng, Ben Thanh, District 1, Ho Chi Minh City",
      "location": "Ho Chi Minh City",
      "local_currency_price_min": 30000.0,
```

Le résultat – Accommodation Planning

```
{
  "accommodations": [
    {
      "name": "Meander Saigon Hostel",
      "description": "A highly-rated hostel with a vibrant atmosphere, great for socializing with other",
      "location": "Ho Chi Minh City",
      "local_currency_price_min": 200000.0,
      "local_currency_price_max": 400000.0,
      "euro_currency_price_min": 6.92,
      "euro_currency_price_max": 13.84,
      "review_score_on_5": 4.4
    },
    {
      "name": "Khoi Hostel",
      "description": "Known for its friendly staff and cozy environment, this hostel provides a welcom",
      "location": "Ho Chi Minh City",
      "local_currency_price_min": 180000.0,
      "local_currency_price_max": 350000.0,
      "euro_currency_price_min": 6.24,
```

Bilan intermediaire

- On a des données specifique pour chaque poste
- Résultat fournis en data mapping exploitable
- Les agents on malgré tout profité les uns les autres de leurs résultats
- Le LLM à accès à un source de données à jour en temps réel
- Mais au fait, c'est toujours pas personnalisé 😞

Les custom tools

Pour personnalisé mon voyage je peux

- ajouter des variables d'entrées dans les prompts, mais ca contraindrait le LLM
- ajouter un outil qui vont lui permettre de me poser des questions

Mais cet outils n'existe pas 🙄

Ca doit être un enfer de créer un outils !

```
class UserInputToolInput(BaseModel): ...

class UserInputTool(BaseTool):
    name: str = "user_input_tool"
    description: str = "Ask a question to the user and return the answer."
    args_schema: Type[BaseModel] = UserInputToolInput

    def _run(self, query: str):
        if not query:
            raise ValueError("Query cannot be empty.")

        answer = input(f"{query}\n\nRéponse : \n")

        if not answer:
            raise ValueError("Answer cannot be empty.")

        return answer
```

Trop simple, on en fait d'autre ?

```
class PassportToolInput(BaseModel):
    name: str = Field(description="Name of passenger")

class PassportTool(BaseTool):
    name: str = "passport_tool"
    description: str = "Read passport from an image."
    args_schema: Type[BaseModel] = PassportToolInput

    def _run(self, name: str):
        file = f"input/{name}.jpg"
        # Check if the file exists
        if not os.path.exists(file):
            raise FileNotFoundError(f"The file {file} does not exist.")

        mrz = read_mrz(file, save_roi=True)

        return {
            "date_of_birth": self.format_date(mrz.date_of_birth),
            "expiration_date": self.format_date(mrz.expiration_date),
            "nationality": mrz.nationality,
            "sex": mrz.sex
        }
```

Démo time



Le résultat – Macro Planning

```
{
  "ordered_cities": [
    {
      "name": "Hanoi",
      "transport_to_come": "Flight from France",
      "euro_transport_price_min": 600.0,
      "euro_transport_price_max": 1000.0
    },
    {
      "name": "Ho Chi Minh City",
      "transport_to_come": "Domestic flight from Hanoi",
      "euro_transport_price_min": 50.0,
      "euro_transport_price_max": 150.0
    },
    {
      "name": "Hoi An",
      "transport_to_come": "Domestic flight to Da Nang, then taxi",
      "euro_transport_price_min": 50.0,
      "euro_transport_price_max": 150.0
    }
  ]
}
```

Le résultat – Activity Planning

```
{
  "activities": [
    {
      "name": "Hanoi Cooking Class & Market Tour",
      "description": "Join a small-group cooking class that starts with a guided market tour to pick f",
      "duration": 4,
      "location": "Hanoi",
      "local_currency_price_min": 800000.0,
      "local_currency_price_max": 1200000.0,
      "euro_currency_price_min": 27.43,
      "euro_currency_price_max": 41.14,
      "review_score_on_5": 4.8
    },
    {
      "name": "Kayaking in Ha Long Bay",
      "description": "Experience the breathtaking beauty of Ha Long Bay while kayaking through emerald",
      "duration": 6,
      "location": "Ha Long Bay",
      "local_currency_price_min": 1500000.0,
```

Le résultat – Food Planning

```
{
  "foods": [
    {
      "name": "Bánh mì Huỳnh Hoa",
      "description": "Famous for its meat-filled baguette with layers of various meats, pâté, and fresh",
      "address": "26 Lê Thị Riêng, Ben Thanh, District 1, Ho Chi Minh City, Vietnam",
      "location": "Ho Chi Minh City",
      "local_currency_price_min": 30000.0,
      "local_currency_price_max": 30000.0,
      "euro_currency_price_min": 1.4,
      "euro_currency_price_max": 1.4,
      "review_score_on_5": 4.8
    },
    {
      "name": "Bánh mì Hồng Hoa",
      "description": "A popular breakfast spot serving crusty baguettes with pork, clever salads, and a",
      "address": "62 Nguyễn Văn Tráng, District 1, Ho Chi Minh City, Vietnam",
      "location": "Ho Chi Minh City",
      "local_currency_price_min": 17000.0,
```